SARA Ben Lashihar

BIM Portfolio 2026



MONZA FS Station Development , 2020

My BIM Journey

2005
B. Sc. Thesis

2006
Development of NCB Cafeteria

2008
UIA Info Point

2009
M. Sc. In Digital Architecture

2009 - 2011
Prada Tokyo Shop & Office

2014
Joddaiem Villas

2015
AinZara Villa Renovation

2016
Remodeling - AutodeskFusion 360

2017
Cooper Residence Remodeling

2019 - 2020
Professional Master in BIM

2020 - 2024
Ph.D. Study

2025
ARCHEIAS Studio

ABOUT

Sara Ben Lashihar is a Postdoctoral Researcher at Maynooth University and the Innovation Value Institute (IVI), specializing in Heritage Building Information Modelling (HBIM), Scan-to-BIM workflows, and digital heritage conservation.

Sara holds a Ph.D. from the Università degli Studi di Pavia (Italy), a Professional Level II Master's degree in Building Information Modelling (BIM) from Politecnico di Milano and the University of Ljubljana, and an M.Sc. in Digital Architecture from the University of Nebraska-Lincoln.

Combining academic research with professional practice, Sara has worked as a BIM Architect in Milan and previously served as an architect and university lecturer in Libya. Sara is also the founder of Alashhar for Arts & Architecture, an initiative providing architectural design services and training in digital design technologies.



Prada Tokyo Shop & Office (2010 – 2016)

Remodeling Prada Tokyo building using Revit Architecture software.

ARCHEIAS Studio



PORDENONE Project, Milan, Italy (2025)

I undertook a comprehensive remodeling of the Revit file, which was originally created using Revit and based on a point cloud survey. I addressed and corrected all identified issues within the model, ensuring greater accuracy and alignment with the existing conditions. In addition, I integrated and developed the new design phase, maintaining consistency with project requirements. Finally, I produced high-quality renderings to effectively visualize the updated design.

CARDUCCI Project, Milan, Italy (2024)

I joined the team in 2024 and contributed to the laser scanning of the building. I was primarily responsible for modeling the existing state based on the point cloud data, completing approximately 70% of the overall model. Additionally, I developed and integrated the proposed renovation phase into the BIM model, ensuring alignment with project goals.

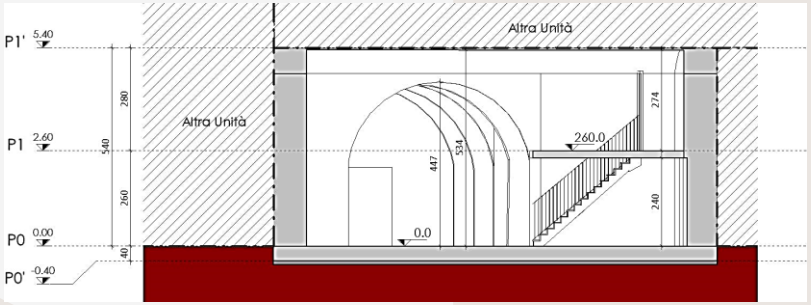


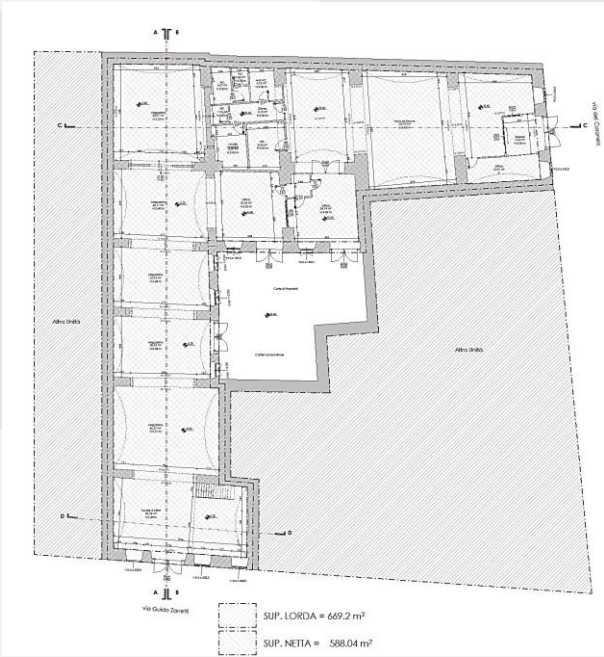
ARCHEIAS Studio

Warehouse Renovation, Trieste, Italy (2025)

I was entrusted with the task of modeling a warehouse is located on the ground floor of a historic building in Trieste. The space, distinguished by its vaulted ceilings, was comprehensively modeled using BIM methodology and based on point cloud data, to support and facilitate future restoration phases.



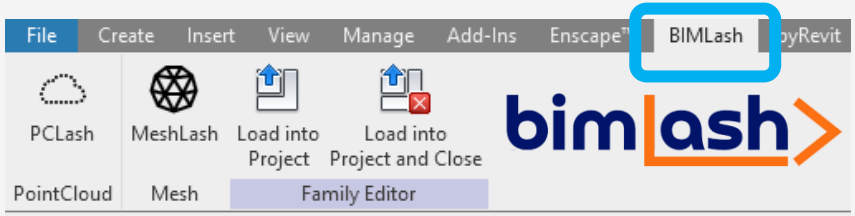




PhD Dissertation (2020 – 2024)

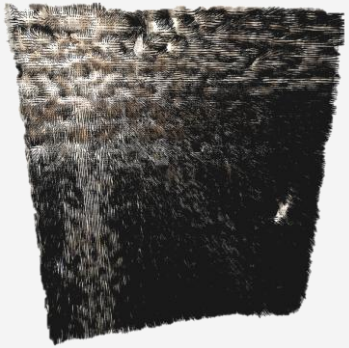
Development and Application of an Intelligent Modeling Process for Heritage Masonry Structures in BIM Applications

The dissertation developed an approach fully implemented in a BIM environment, the **Automatic Python Workflow (APW).** This workflow is primarily based on the Python language, which converting any point cloud of a masonry element into a Revit family. A Revit panel named “BIMLash” has been created to simplify the extensive approach into a more concise format.





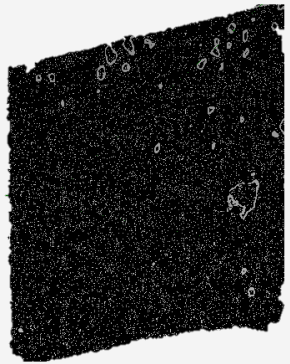
Original Point Cloud
(Python + Open3D)



Calculation of Normals
(Python + Open3D)



The generated mesh
in Dynamo



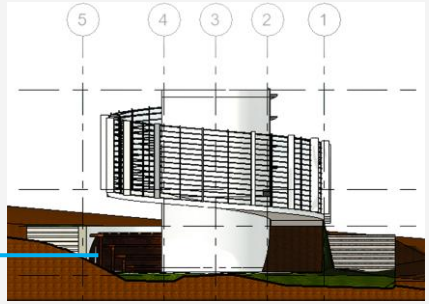
The generated mesh
in Revit Family

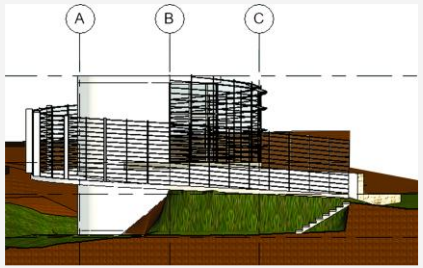
Professional Master in BIM



MONZA FS Station Development , 2020

This research developed the connectivity solutions for the area around the Monza FS Railway station and the architectural design of the promised new metro station. The new Monza FS metro station is an essential part of the new extension of the of the Milan Metro Line 5 project.







Italian Pavilion, Guimaraes, Portugal, 2019

A comprehensive proposal for the Italian pavilion at the upcoming international exhibition in Portugal. The design concept aims to encapsulate the essence of Italian culture and hospitality, providing an inviting space for visitors from around the world.

Remodeling





Cooper Residence - Remodeling (2017)

"Cooper Residence " The building was designed by Gwathmey & Siegel in 1969. The residence is located in Orleans, Massachusetts.





Remodeling - Fusion360 (2016)

"Moroccan Sink" It is a tutorial for Fusion 360 online course.

Designing

AinZara Villa Renovation, (2015)

A renovation project for a small house in AinZara area that was built with customary style in 2009. The new design was based on adding new level with Mediterranean-style. Autodesk Revit Architecture 2015 was employed in the design process of the house which was started in March 2015.





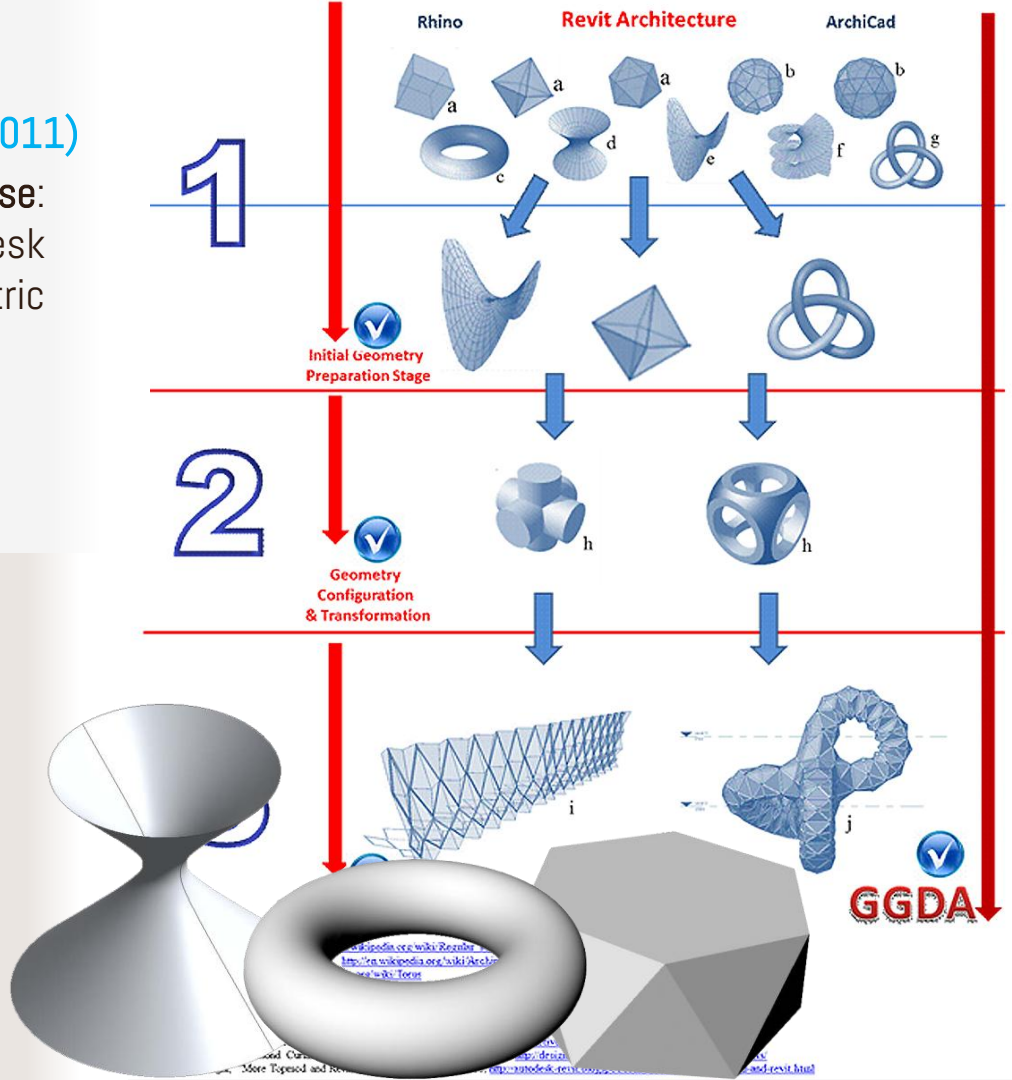
Joddaiem Villas (2014)

A design project in Joddaiem at west side of Tripoli, consists of two identical villas and a multipurpose hall. The project is a part of my work in the proposed research for sustainable development. The research employs the technology of Building Information Modeling (BIM) and energy analysis applications in the redesign process of the Tripolitan House.

M. Sc. In Digital Architecture (2009 – 2011)

BIM Applications in Conceptual Design Phase: Exploring the Abilities and Limitations of Autodesk Revit Architecture through the Generative Geometric Design Approach.

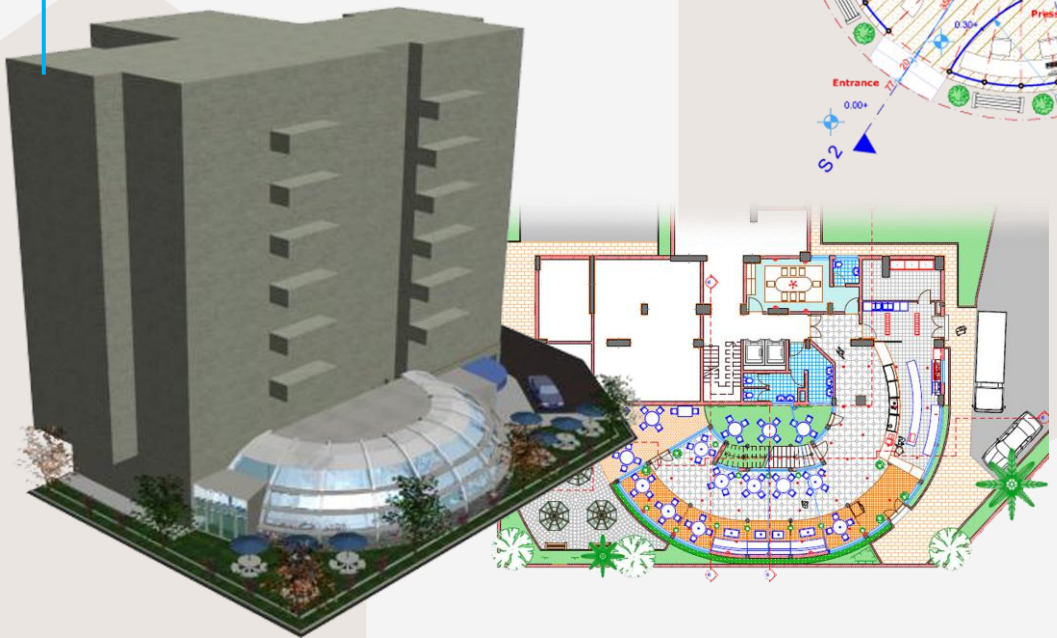
This thesis examines how Autodesk Revit Architecture supports the Generative Geometric Design Approach (GGDA), a method that emphasizes the use of geometric forms during the conceptual design phase. Through a multi-method research strategy, the study investigates both the strengths and limitations of Revit in applying GGDA. Findings reveal that while Revit offers some useful capabilities for early-stage generative design, it also has clear constraints. The research highlights the importance of understanding these limitations for both software development and architectural education, and it encourages further studies to explore the compatibility of BIM tools with generative design approaches.

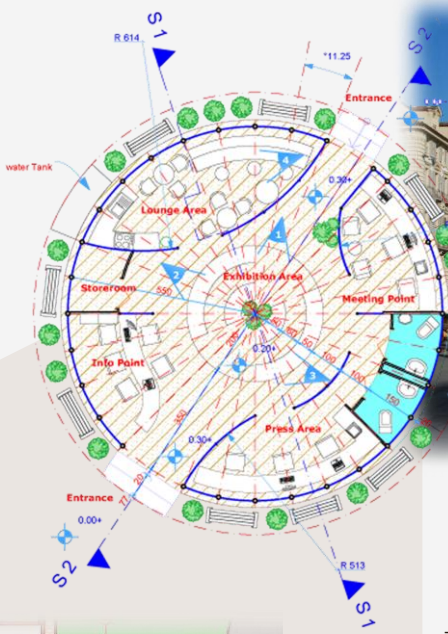


Competitions

Development of NCB's Cafeteria Competition 1st Place , (2006)

It was an architectural design competition for intern-architects who work in NCB. The purpose of the competition is to expand the existing cafeteria to contain 120 employees as maximum.







UIA INFO POINT Competition Torino, Italy (2008)

The goal of the competition is to design an info point for UIA World Congress in a plaza in Torino. The design should be focused on the overall theme of the congress "Transmitting Architecture". The theme is based on the idea of horizontal, democratic urban space in which people can meet, offer and get information.

Sara Ben Lashihar



It would be a pleasure to speak with you and discuss the experience I can bring to your firm. I can be reached via email at sara.benlashihar@mu.ie or by phone at +353 0830057491, and I would be glad if you could take the time and give me a chance to speak more about my skills.